

SEQUENCES & SERIES

What is a Series?

A Series is the sum of terms of a Sequence.

Example:

$2+4+6+8+10+\dots$ is a series.

* $1+3+5+7+9+\dots$

* $5+10+15+20+\dots$

* $2+4+8+16+32+64+\dots$

What is a Sequence?

A Sequence is a list of numbers written in a special order.

Example:

$2, 4, 6, 8, 10, \dots$ is a sequence [even numbers]

$1, 3, 5, 7, 9, \dots$ [odd numbers]

* $5, 10, 15, 20, 25$

* $2, 4, 8, 16, 32$

Types of Sequences -

① Arithmetic Progression (A.P.)

↳ A sequence where the difference between any two terms is constant

Example -

$5, 8, 11, 14, \dots$ (common difference $d=3$)

② Geometric Progression (G.P.)

↳ A sequence where each term is multiplied by the same number (common ratio).

Example -

$3, 6, 12, 24, \dots$ (common ratio $r=2$).

③ Harmonic Progression (H.P.)

↳ A sequence where reciprocals of terms form an A.P.

Example -

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$

Take reciprocals $\rightarrow 1, 2, 3, 4 \rightarrow$ A.P

④ Special Series

$$1+2+3+\dots+n = \frac{n(n+1)}{2}$$

$$1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3+2^3+3^3+\dots+n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

⑤ Sequences:

* Arithmetic Progression (A.P.)

$2, 4, 6, 8, 10$
 $+2 \quad +2 \quad +2 \quad +2$

Any 2 term की difference एक जैसा (constant)

Hack $\rightarrow +/-$

* Geometric Progression (G.P.)

$2, 4, 8, 16, 32, 64$
 $\times 2 \quad \times 2 \quad \times 2 \quad \times 2 \quad \times 2$

same no.

$\Rightarrow$ multiply करते जाइए

Hack $\rightarrow \times / \div$

* Harmonic Progression (H.P.)

$\Rightarrow$ A.P. उसका reciprocal

$\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{10}$

$\uparrow$ $2, 4, 6, 8, 10$
Reciprocal [A.P.]

8. Write the first five terms of each of the Sequences n th terms are:

① $a_n = n(n+2)$

1st term

$(n=1)$

$a_1 = 1(1+2)$

$a_1 = 1(3)$

$a_1 = 3$

2nd term

$(n=2)$

$a_2 = 2(2+2)$

$a_2 = 2(4)$

$a_2 = 8$

3rd

$(n=3)$

$a_3 = 3(3+2)$

$a_3 = 15$

4th

$(n=4)$

$a_4 = 4(4+2)$

$a_4 = 24$

5th

$(n=5)$

$a_5 = 5(5+2)$

$a_5 = 35$

Example -

What is the 20th term of the sequence defined by.

$$(n=20) \quad a_n = (n-1)(2-n)(3+n)?$$

$$a_{20} = (20-1)(2-20)(3+20) \\ = (19)(-18)(23)$$

$$\boxed{a_{20} = -7866}$$

Example -

Let the sequence (a_n) be defined as follows, find 1st 5 terms.

$$a_1 = 1, \quad a_n = a_{n-1} + 2 \quad \text{for } n \geq 2$$

$n=2$	$n=3$	$n=4$	$n=5$
$a_2 = a_{2-1} + 2$	$a_3 = a_{3-1} + 2$	$a_4 = a_{4-1} + 2$	$a_5 = a_{5-1} + 2$
$= a_1 + 2$	$= a_2 + 2$	$= a_3 + 2$	$= a_4 + 2$
$(\because a_1 = 1)$	$= 3 + 2$	$= 5 + 2$	$= 7 + 2$
$= 1 + 2$	$\boxed{a_3 = 5}$	$\boxed{a_4 = 7}$	$\boxed{a_5 = 9}$
$\boxed{a_2 = 3}$			

Question) $a_n = 2^n$

1st 5 terms, $a_{30} = ?$, $a_{25} = ?$

$$a_1 = 2, \quad a_4 = 16$$

$$a_2 = 4, \quad a_5 = 32$$

$$a_3 = 8$$

$$\boxed{a_{30} = 2^{30}}$$

$$\boxed{a_{25} = 2^{25}}$$

Question) Find the indicated terms in each of the sequences in Exercise 7 to 10 whose n th terms are:

$$(i) \quad a_n = \frac{n^2}{2n}; \quad a_7$$

$$a_7 = \frac{7^2}{2 \times 7}$$

$$= \frac{49}{14}$$

$$\boxed{a_7 = 7/2}$$

Ans.

Geometric Progression (G.P.) -

A Geometric Progression (G.P.) is a sequence of numbers where each term (except the first) is obtained by multiplying the previous terms by a fixed number called the common ratio (r).

Examples:

$$1. \quad 2, 4, 8, 16 \rightarrow (r=2)$$

$$2. \quad 1, 1/3, 1/9, 1/27 \dots \rightarrow (r=1/3)$$

$$3. \quad 0.01, 0.0001, 0.000001 \rightarrow (r=0.01)$$

A.P	G.P
$2, 4, 6, 8, 10$ $+2 \quad +2 \quad +2 \quad +2$ common difference (d) = 2	$2, 4, 8, 16, 32, 64$ $\times 2 \quad \times 2 \quad \times 2 \quad \times 2 \quad \times 2$ common ratio (r) = 2

Exponent Rules

For $a \neq 0, b \neq 0$

Product Rule	$a^x \times a^y = a^{x+y}$
Quotient Rule	$a^x \div a^y = a^{x-y}$
Power Rule	$(a^x)^y = a^{xy}$
Power of a Product Rule	$(ab)^x = a^x b^x$
Power of a Fraction Rule	$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$
Zero Exponent	$a^0 = 1$
Negative Exponent	$a^{-x} = \frac{1}{a^x}$
Fractional Exponent	$a^{\frac{x}{y}} = \sqrt[y]{a^x}$

General Forms of G.P.:

A Geometric Progression (G.P) is a sequence of numbers where each term (except the first) is obtained by multiplying the previous term by a fixed number called the common ratio (r).

If;

a = first term
 r = common ratio

Then G.P. is -

$$a, ar, ar^2, ar^3, \dots, ar^{n-1}$$

Find T_{20} :

$$T_{20} = a \cdot r^{20-1}$$

$$T_{20} = 2 \cdot 2^{19}$$

$$= (2^{20})$$

final Answer.

Formula: $T_n = a \cdot r^{n-1}$

Example -

Find 4th term of G.P 2, 4, 8, ...

$$T_4 = 2 \times (2)^3 = 16$$

Sum of G.P.

$$\text{Let } S_n = a + ar + ar^2 + \dots + ar^{n-1}$$

Case ①: If $r=1$

$$\text{Then } S_n = na$$

Case ②: If $r \neq 1$.

$$\text{Then } S_n = a \times (r^n - 1) / (r - 1)$$

'CR'

$$S_n = a \times (1 - r^n) / (1 - r) \text{ [if } r < 1]$$

G.P

sum

$r > 1$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

A.P

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

A.P

$$T_n = a_n = a + (n-1)d$$

a = First term

d = common difference

n = n th terms.

G.P

$$T_n = a_n = a \cdot r^{n-1}$$

a = first term

r = common ratio

n = n th term.

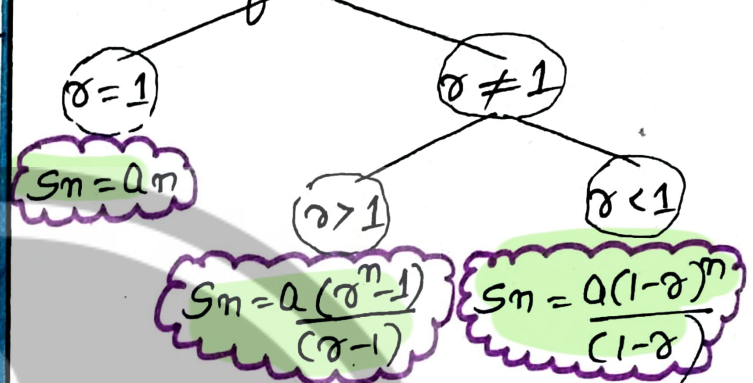
Infinite G.P sum (When $|r| < 1$)

If the terms go on forever and $|r| < 1$:

$$S_{\infty} = a / (1 - r)$$

$$\oplus S_{\infty} = \frac{a}{1 - r}$$

Sum of 'n' Terms.



Quantity	Formula
n^{th} term	$T_n = a \times r^{n-1}$
Sum of n terms (r not equal 1)	$S_n = a \times (r^n - 1) / (r - 1)$
Sum if $r = 1$	$S_n = na$
Sum to infinity	$S_{\infty} = a / (1 - r)$ if

Geometric Mean (G.M).

The Geometric Mean (G.M) of two positive numbers a and b is:

$$G.M = \sqrt{ab}$$

$$\therefore 2, 4, 8, 16, 32$$

Example:

G.M of 2 and 8

$$= \sqrt{(2 \times 8)} = \sqrt{16} = 4$$

$$\text{G.M } 4 \text{ \& } 16$$

$$G.M = \sqrt{ab} \\ = \sqrt{4 \times 16} \\ = \sqrt{64}$$

We see: 2, 4, 8 are in Geometric Progression.

$$\boxed{G.M = 8}$$

Example) Find the 20th & n th terms of the G.P $\frac{5}{2}, \frac{5}{4}, \frac{5}{8}, \dots$

$$T_n = ar^{n-1}$$

$$T_{20} = ar^{19}$$

$$T_{20} = \left(\frac{5}{2}\right) \left(\frac{1}{2}\right)^{19}$$

$$= 5 \times \left(\frac{1}{2}\right)^{19} \times \left(\frac{1}{2}\right)^{19}$$

$$= 5 \times \left(\frac{1}{2}\right)^{1+19}$$

$$T_n = 5 \times \left(\frac{1}{2}\right)^{20} \text{ Ans.}$$

$$T_n = ar^{n-1}$$

$$T_n = \left(\frac{5}{2}\right)\left(\frac{1}{2}\right)^{n-1}$$

$$= 5 \times \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^{n-1}$$

$$= 5 \times \left(\frac{1}{2}\right)^{1+n-1}$$

$$T_n = 5 \times \left(\frac{1}{2}\right)^n \text{ Ans.}$$

$$a = \frac{5}{2}$$

$$r = \frac{5/4}{5/2} = \frac{5}{4} \times \frac{2}{5} = \frac{1}{2} \therefore r = \frac{1}{2}$$

Concept	
G.P	A.P
$T_n = ar^{(n-1)}$ 20th $\Rightarrow ar^{20-1}$ $\hookrightarrow ar^{19}$ * 30th $\Rightarrow ar^{29}$ * 150 $\Rightarrow ar^{49}$ * 10 $\Rightarrow ar^9$	20th Term $a_n = a + (n-1)d$ $a_{20} = a + (20-1)d$ $a_{20} = a + 19d$ * $a_{50} = a + 49d$ * $a_{10} = a + 9d$

Example) Find the 12th term of a G.P whose 8th term is 192 and the common ratio is 2.

$$T_{12} = ?$$

$$T_{12} = ar^{11}$$

$$T_{12} = \left(\frac{3}{2}\right)(2)^{11}$$

$$= 3 \times \frac{1}{2} \times 2^{11}$$

$$= 3 \times 2^{10} \text{ Ans.}$$

$$\left[\frac{a^m}{a^n} = a^{m-n}\right]$$

Example) The 5th, 8th and 11th terms of a G.P are P, Q and S, respectively. Show that $Q^2 = PS$.

$$(ab)^m = a^m \times b^m$$

$$(a^m)^n = a^{m \times n}$$

$$T_5 = P \quad T_8 = Q \quad T_{11} = S$$

$$ar^4 = P \quad ar^7 = Q \quad ar^{10} = S$$

Show that $\Rightarrow Q^2 = PS$

$$(ar^7)^2 = (ar^4)(ar^{10})$$

$$a^2(r^7)^2 = a \cdot r^4 \cdot a \cdot r^{10}$$

$$a^2(r)^{7 \times 2} = a^{1+1} \cdot r^{4+10}$$

$$a^2 r^{14} = a^2 \cdot r^{14}$$

LHS = R.HS (Hence Proved).

Imp

Example) For what value of x, the numbers $-\frac{2}{7}, x, -\frac{7}{2}$ are in G.P?

$$\frac{-A}{B} = \frac{-A}{B} = \frac{A}{-B}$$

$$a_1, a_2, a_3$$

Common Ratio:

$$r = \frac{a_2}{a_1}$$

$$r = \frac{x}{-2/7}$$

$$r = \frac{x}{1} \times \frac{7}{-2}$$

$$r_1 = \frac{-7x}{-2} \quad \text{--- (1)}$$

$$r = \frac{a_3}{a_2}$$

$$r = \frac{-7/2}{x}$$

$$r = \frac{-7}{2} \times \frac{1}{x}$$

$$r_2 = \frac{-7}{2x} \quad \text{--- (2)}$$

Common ratios are same in G.P.

$$\hookrightarrow r_1 = r_2$$

$$\Rightarrow \frac{-7x}{-2} = \frac{-7}{2x}$$

$$\Rightarrow x = \frac{1}{x}$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm \sqrt{1}$$

$$x = \pm 1$$

Example) Find the sum to indicated numbers of terms in each of the Geometric Progression $\sqrt{7}, \sqrt{21}, 3\sqrt{7}, \dots$

n terms

$$a = \sqrt{7}$$

$$r = \frac{\sqrt{21}}{\sqrt{7}}$$

$$= \sqrt{21/7}$$

$$r = \sqrt{3}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{\sqrt{7}[(\sqrt{3})^n - 1]}{\sqrt{3} - 1}$$

solve yourself

If sum of three terms and their product is given:

Letting 3 Terms in A.P. -

Let the three terms be:

$$\{a-d, a, a+d\}$$

Properties:

Middle Term is average of others.

$$\text{Sum} = 3a$$

$$\text{Common difference} = d$$

Letting 3 Terms in G.P. -

Let the three terms be:

$$\{ \frac{a}{r}, a, ar \}$$

Properties:

Each term is multiplied/divided by common ratio r

$$\text{Product} = a^3$$

$$\text{Common Ratio} = r$$

Imp

Example)

How many terms of G.P $3, 3^2, 3^3, \dots$ are needed to give the sum 120?

$$a = 3 \quad | \quad S_n = \frac{a(r^n - 1)}{r - 1}$$

$$r = \frac{3^2}{3} \quad | \quad 120 = \frac{3(3^n - 1)}{3 - 1}$$

$$r = 3 \quad | \quad \frac{120 \times 2}{3} + 1 = 3^n$$

$$r > 1 \quad | \quad (n = 4) \quad \text{Ans.}$$

Imp

Example)

Given a G.P with $a = 729$ and 7th term 64 determines S_7 .

$$\begin{aligned} r &= \frac{2}{3} \\ r &= 0.66 \\ r &< 1 \end{aligned} \quad \left| \quad \begin{aligned} T_7 &= 64 \\ a r^6 &= 64 \\ 729 r^6 &= 64 \\ r^6 &= 64/729 \end{aligned} \right. \quad \begin{aligned} r^6 &= 2^6/3^6 \\ r^6 &= (\frac{2}{3})^6 \\ r &= \frac{2}{3} \end{aligned}$$

$$r < 1 \rightarrow S_7 = \frac{a(1-r^n)}{(1-r)}$$

$$= \frac{729[1-(\frac{2}{3})^7]}{1-\frac{2}{3}}$$

calculate yourself -

Example)

If the 4th, 10th and 16th terms of a G.P are x, y and z , respectively.

Prove that x, y, z are in G.P.

check x, y, z are in A.P

$$T_4 = x$$

$$a r^3 = x$$

$$T_{10} = y$$

$$a r^9 = y$$

$$T_{16} = z$$

$$a r^{15} = z$$

$$a r^3, a r^9, a r^{15}$$

common Ratio

$$r = \frac{a r^9}{a r^3}$$

$$r = r^6$$

common Ratio

$$r = \frac{a r^{15}}{a r^9}$$

$$r = r^6$$

yes in G.P

Example)

If a, b, c and d are in G.P. show that $(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$

$$\begin{matrix} a & b & c & d \\ \downarrow & \downarrow & \downarrow & \downarrow \\ T_1 & T_2 & T_3 & T_4 \end{matrix}$$

$$a \quad ar \quad ar^2 \quad ar^3$$

$$\begin{aligned} (a^2 + (ar)^2 + (ar^2)^2) & ((ar)^2 + (ar^2)^2 + (ar^3)^2) = \\ [a(ar) + ar(ar^2) + ar^2(ar^3)]^2 & \\ \Rightarrow (a^2 + a^2 r^2 + a^2 r^4)(a^2 r^2 + a^2 r^4 + a^2 r^6) & = \\ [a^2 r + a^2 r^3 + a^2 r^5] & \end{aligned}$$

$$\Rightarrow a^2(1+r^2+r^4) a^2 r^2(1+r^2+r^4) = [a^2 r(1+r^2+r^4)]^2$$

$$\Rightarrow (1+r^2+r^4)^2 = (a^2 r)^2 (1+r^2+r^4)^2 = a^4 r^2 (1+r^2+r^4) \quad \text{Ans}$$

Name in Latin	Theorem
Alterando	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$
Invertendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$
Componendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{b} = \frac{c+d}{d}$
Dividendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a-b}{b} = \frac{c-d}{d}$
Componendo et Dividendo	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$

Example

The sum of two numbers is 6 times their geometric mean, show that numbers are in the ratio $(3+2\sqrt{2}) : (3-2\sqrt{2})$.

$$a+b = 6\sqrt{ab}$$

$$a+b = 3 \times 2\sqrt{ab}$$

$$\frac{a+b}{2\sqrt{ab}} = 3$$

By comp & dividendo rule.

$$\Rightarrow \frac{(\sqrt{a})^2 + (\sqrt{b})^2 + 2\sqrt{ab}}{(\sqrt{a})^2 + (\sqrt{b})^2 - 2\sqrt{ab}} = \frac{3}{1}$$

$$\Rightarrow \frac{(\sqrt{a} + \sqrt{b})^2}{(\sqrt{a} - \sqrt{b})^2} = \frac{3}{1}$$

$$\Rightarrow \left(\frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} \right)^2 = \frac{3}{1}$$

$$\Rightarrow \frac{\sqrt{a} + \sqrt{b}}{\sqrt{a} - \sqrt{b}} = \frac{\sqrt{3}}{1}$$

Again By C&D Rule:

$$\Rightarrow \frac{\sqrt{a} + \sqrt{b} + \sqrt{a} - \sqrt{b}}{\sqrt{a} + \sqrt{b} - (\sqrt{a} - \sqrt{b})} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$\Rightarrow \frac{\sqrt{a} + \sqrt{a}}{\sqrt{b} + \sqrt{b}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

Squaring

both sides:

$$\frac{(\sqrt{a})^2}{(\sqrt{b})^2} = \frac{(\sqrt{3} + 1)^2}{(\sqrt{3} - 1)^2}$$

$$\Rightarrow \frac{a}{b} = \frac{(\sqrt{3})^2 + (1)^2 + 2\sqrt{3}}{(\sqrt{3})^2 + (1)^2 - 2\sqrt{3}}$$

$$= \frac{3 + 1 + 2\sqrt{3}}{3 + 1 - 2\sqrt{3}}$$

$$\frac{a}{b} = \frac{3 + 2\sqrt{3}}{3 - 2\sqrt{3}}$$

Relation Between A.M and G.M

Let two positive numbers be a and b .

Arithmetic Mean (A.M):

$$A = \frac{a+b}{2}$$

Geometric Mean (G.M):

$$G = \sqrt{ab}$$

Relationship

Let's compare A & G .

$$A - G = \frac{a+b}{2} - \sqrt{ab}$$

Let's simplify.

$$A - G = \frac{a+b-2\sqrt{ab}}{2}$$

This is same as:

$$A - G = \frac{(\sqrt{a} - \sqrt{b})^2}{2}$$

Since squares are always ≥ 0 .

$$(\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$\text{So, } A - G \geq 0$$

Final Result: $A.M \geq G.M$

Equality holds when $a=b$.